DRY RUN

Product and engineering brief

Prepared October 9, 2026. Initial development target: a coordinated three-hour first-review build, followed by student implementation, device validation, content review, and frontend refinement.

Dry Run helps families rehearse the first minutes and next 72 hours of a disaster, understand the consequences for their household, and keep a practical plan available offline.

The strongest product combines three things: an explainable household dependency model, a user-confirmed 3D home with practice routes, and a short list of preparations whose effects can be demonstrated by replaying the same exercise. The complete digital exercise stays on the phone. Real emergencies and real-home verification can require actions beyond the phone.

Recommended decisions

Area	Decision
Product focus	House-fire rehearsal and power-failure cascades are the two flagship demonstrations. All six hazards share a real engine and have distinct rules.
Apple stack	SwiftUI, RoomPlan, RealityKit, MapKit, local PDF generation, and an encrypted on-device vault.
Deployment target	iOS 18 baseline; independently gate LiDAR capture and optional on-device Foundation Models.
Backend	A small public-data and content service. Household profiles, scans, simulations, priorities, and private plans remain on the phone.
Initial review	A coherent working prototype, reproducible fixtures/tests, precise feature status, and an honest AI-contribution record.

1. Positioning and competition constraints

The central question is personal and concrete: 'If this fails, what stops working in our home, and what should we prepare first?' A general disaster-information app can tell a family to store supplies. Dry Run can explain that its electric well pump, heating controls, and essential equipment depend on the same power source, and show what changes when one preparation is added.

Use a truthful local pitch. Texas's current 25th District includes all or parts of 13 counties, and the ACS 2024 one-year estimate is 826,421 residents. That population describes the community; it is not an estimate of accessible devices, users reached, or lives protected. The iOS-only decision makes manual layouts and printable plans especially valuable. Confirm the applicable district for competition eligibility instead of assuming that a county or ZIP is wholly in TX-25. [S03, S04]

Texas DSHS confirmed 246 deaths related to the February 2021 winter storm. These were statewide storm-related deaths, not all caused specifically by power loss. The state-climatologist assessment described by Texas A&M projects growing extreme heat, drought, wildfire-condition, and urban-flooding risks through 2036. Avoid a blanket statement that every included hazard will worsen over the next decade. [S05, S06]

Past CAC winners already describe individualized disaster plans, shelter discovery, and evacuation features. DisasterReady is one documented example. The defensible distinction is the combination Dry Run demonstrates: causal household modeling, practice inside a verified home, visible before-and-after comparisons, and useful offline outputs. Targeted research cannot establish that no prior winner has ever used a similar component. [S07]

The 2026 CAC allows disclosed AI support but requires meaningful student technical contributions and understanding. AI cannot constitute the entirety of technical development. Treat the autonomous build as a reference prototype and foundation, preserve an assistance ledger, and plan substantial student implementation before submission. The national deadline is October 26 at noon Eastern, or 11 a.m. Central. Use that cutoff despite a later time advertised on the district page. [S01, S02]

The user should own a substantive technical contribution: for example, improving scan-to-graph extraction, implementing and testing an accessible route mode, designing a transparent prioritization method, or coding and testing the local information ingestion/cache/freshness pipeline. Reviewing generated code and changing its colors should not be treated as sufficient proof of contest compliance.

2. The complete household journey

Start with a fictional demonstration household so the app is useful before the user completes setup. Real household onboarding should take a few minutes, then ask follow-up questions only when they affect a result. Allow 'not sure' throughout.

Information to collect

Group	Useful inputs
People and pets	Aliases; broad age bands; assistance or communication needs; sleeping rooms; pets and carriers; optional helper assignments.
Home	House/apartment/manufactured home; floors; tenancy constraints; heating/cooling type and electrical dependencies; water and sanitation setup.
Essential equipment	Purpose, required power, manufacturer-confirmed backup compatibility/runtime if known, storage requirements, and local care-plan notes.
Transport and support	Car or no car; seats/accessibility; fuel or charging assumptions; agreed support contacts; feasible destination and transport plan.
Supplies	Stored safe water, useful food/storage, lights, batteries, verified backups, and budget/storage constraints.
Area	Manually selected county/region; optional precise address retained locally; selected official contacts.

Do not require legal names, dates of birth, diagnoses, identity documents, or a full medical history. Ask about functional needs and dependencies. A household can complete an exercise with unknowns; the result should identify what those unknowns prevent the app from assessing.

The full flow is household setup, home capture/manual layout, confirmation, scenario selection, rehearsal decisions, debrief, ranked preparations, same-scenario replay, and an offline plan. Region context and the help directory support this flow. They should not turn the home screen into an emergency-news dashboard.

Distinguish 'Try this preparation' from 'We have completed this preparation.' A simulated battery or transport arrangement must never silently become a real item in the household's inventory.

3. Two clocks and six distinct scenarios

The 72-hour clock is valuable for outages, supplies, and recovery. Home fire, tornado, and flash flood also require immediate decisions. USFA notes that people may have less than two minutes to escape once an alarm sounds; NWS directs prompt sheltering or movement away from rising floodwater. The app should use an immediate phase and then, where useful, a longer recovery phase. [S19, S22, S23]

Scenario	Primary rehearsal	Essential modeling distinction
Freeze + power loss	Heating, equipment continuity, water, food, communications, transport, warmer destination.	Gas heating may still require electricity. A well and municipal supply behave differently.
Heat + grid strain	Cooling access, assistance, equipment/storage, transport to an appropriate destination.	Grid strain is not itself an outage. No invented indoor temperature or personal medical deadline.
Tornado	Prepared shelter choice, warning response, assistance, nighttime awareness.	A scanned closet is not a certified safe room; manufactured-home limitations matter.
Flash flood	Early relocation, higher ground, transport and crossing constraints, official instructions.	A map road is not evidence of safe passage; floodwater is a hard exclusion.
Home fire	Exit choice, blocked route, alternative, meeting point, household assistance.	Practice with authored exclusions; do not predict fire spread or safe time remaining.
Water outage	Safe stored supply, advisory type, cooking, hygiene, sanitation, pets/equipment.	Quantity and safety are different. Boil-water, do-not-drink, and do-not-use are distinct.

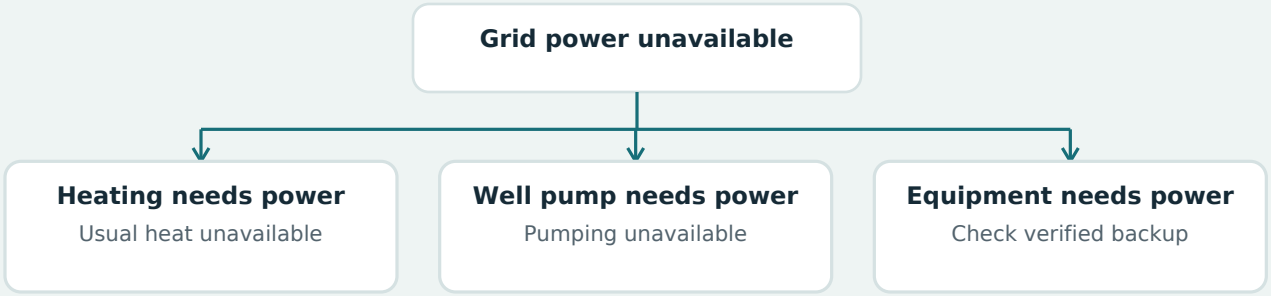
Every exercise needs a few consequential choices and a clear debrief. Use fictional event schedules that the user can inspect. Do not display a demo warning as current or let the simulated clock resemble an official alert timestamp.

For heat, preserve source-based cooling guidance and medical-care boundaries. For freeze, never recommend an oven as home heating or an indoor generator. For flood, exclude flooded crossings instead of merely adding a routing cost. For water contamination, boiling is not a universal remedy. [S20-S24]

4. The cascade model

Use an on-device deterministic discrete-event engine. It processes an immutable household snapshot, a versioned scenario, a stable event schedule/seed, and a set of decisions. Results include service states, resource quantities, unmet requirements, and explanations tied to inputs and source-backed rules.

EXAMPLE: ONE FAILURE, THREE HOUSEHOLD DEPENDENCIES



Apply the household facts. Preserve compatible backup paths and stored supplies.

Explain the dependency. Identify the gap. Replay one preparation.

The example diagram shows possible branches for a household with those dependencies. It does not say all homes lose all three services when the grid fails. Compatible backup paths and confirmed stored resources can interrupt a branch.

Model AND/OR requirements. Equipment may require a working device AND compatible power; grid OR a suitable backup can provide power. Track a shared battery once, even when several loads use it. Prevent negative supplies, duplicate consumption, and unjustified replenishment after restoration.

Keep operational, degraded, unavailable, and unknown states distinct. A well pump stopping does not erase stored water. A municipal outage is a separate event. Losing Wi-Fi does not prove cellular service is down. Track handset charge and compatible charging separately, using declared exercise/runtime assumptions rather than a supposed prediction of the real phone. A powered phone retains offline plans without connectivity; a fictional dead phone affects the scenario while the real app remains usable. Restored electricity does not clear a water advisory or prove a medical device's settings are correct.

Medical-device runtime and backup compatibility should come from instructions, the manufacturer, and the household care plan. Unknown runtime produces a confirmation task, not a made-up countdown. The FDA explicitly advises checking device instructions or the manufacturer regarding batteries and generators. [S25]

Use numerical calculations where the evidence supports them. The CDC's baseline emergency-water storage recommendation is at least one gallon per person per day for three days, with additional needs considered. Four people therefore have a baseline three-day storage target of 12 gallons before extras. This is a supply-planning quantity, not a prescribed individual fluid intake. [S26]

Do not estimate exact indoor temperature, smoke toxicity, injury probability, or survival time from a basic questionnaire and scan. Transparent authored scenarios are already useful without those unsupported predictions.

Evidence and revisions

Every result should retain household/home revisions, scenario and rule versions, regional/locale pack versions, seed, matched input IDs, triggered events, source IDs, and unresolved assumptions. An asynchronous result from an old run cannot overwrite a new household state. When a door, device, or assistance requirement changes, dependent routes and plans need regeneration.

5. Ranked preparations with visible effects

Prioritize unmet essential equipment, assistance, shelter, or escape needs first; then unknown facts that block a dependable plan; then scenario-specific gaps; then preparations useful across several hazards. Use cost, effort, space, renter constraints, and preferences as tie-breakers within appropriate priority bands.

This produces useful no-cost actions as well as supply purchases. Examples include confirming a backup arrangement with the equipment provider, agreeing on transport to a cooled location, verifying an exterior door, assigning help during an exercise, and saving the water provider's contact number.

Each card should explain the action, why it applies to this household, which event exposed the gap, what the model says it improves, and what is still unconfirmed. Use the top three cards in the normal view and keep details available. Avoid a single readiness percentage that implies scientific calibration or encourages purchases to improve a score.

Replay is central to the product. Reconstruct the initial state, apply one intervention, and replay the same external events. Show a structured difference: this dependency remains supported, this supply gap shrinks, or this route no longer depends on the blocked portal. Do not compare two randomly different disasters or carry depleted state into the new run.

6. RoomPlan to a useful home model

Apple's RoomPlan is a strong fit for clean geometric reconstruction. It uses camera and LiDAR data to produce a parametric room representation with dimensions and recognized objects. Gate capture with RoomCaptureSession.isSupported; the rest of Dry Run remains available without a scanner. [S08, S10]

Use RoomCaptureView for acquisition/coaching, RoomBuilder for individual-room processing, and StructureBuilder.capturedStructure(from:) for multiroom assembly. Retain a shared AR coordinate system or use Apple's supported relocalization workflow. Independently exported rooms cannot be positioned reliably by guessing. Apple's published multiroom guidance favors single-floor residential spaces around 2,000 square feet; treat this as a recommended operating envelope, not a guarantee. [S09]

Keep three separate records: local scan evidence, the canonical user-reviewed HomeModel, and derived display/navigation data. The rendered USDZ model is not the source of routing truth. Preserve metric units, floor frames, rotations, stable identifiers, source provenance, and user overrides.

After capture, ask the user to confirm room connections, exterior doors, operable openings, stairs, obstacles/clearances, and the outside meeting place. Recognition of a staircase or window does not prove usability as an exit. Apartment common areas may be outside the scan and need an explicit incomplete-building-plan state.

Offer manual layout creation and a fictional sample home from the first build. The initial editor can use rectangular rooms and explicit door connections; it does not need to rival a CAD application. This supports non-LiDAR phones, partially scanned homes, and credible testing.

Useful annotations are smoke/CO alarms, exits, extinguishers, utility shutoffs, essential equipment, supplies, and the meeting point. Preserve the requested outdoor hydrant marker as a separate landmark. It should not become a household firefighting objective.

Use RealityKit for a restrained dollhouse with cutaway walls, floor switching, room names, annotated items, and routes. A 2D plan and accessible room list should expose the same underlying information. Do not spend the initial window on realistic smoke, cinematic flames, or decorative furniture assets. [S11, S12]

7. Evacuation routing that respects the evidence

Maintain a geometric egress graph separate from the utility-dependency graph. Start with reviewed portals and within-room waypoints. Each traversable edge includes its actual geometric polyline, width/clearance evidence, floor change, operability, verification state, and suitability for the household member.

Apply hard exclusions before Dijkstra/A* ranks remaining candidates by distance or another documented cost. Exclude authored fire/smoke/blockage zones, unknown exterior destinations, incompatible mobility constraints, and paths that cross walls or obstacles. Do not use a high cost as a substitute for exclusion.

A connected room graph can still draw a line through a couch or across the missing corner of an L-shaped room. Validate every displayed segment in the allowed floor geometry. The renderer must use the validated path rather than smooth it independently into an invalid shortcut.

An alternate route should use a different exit or avoid the important chokepoint. Two routes through the same blocked hallway are not independent. Detected windows need confirmation; upper-floor windows are not ordinary walking edges. If the model lacks a verified route, say so and identify why. Never invent an opening to produce a reassuring result.

Use labels such as 'Practice route' and 'Hallway blocked in this exercise.' A scanned layout cannot establish current smoke conditions, future fire behavior, structural integrity, or a universally safest route. Keep actionable fire instructions source-based and reviewed. [S19]

The flagship fire demo should show a household starting in a known room, a kitchen-side route becoming unavailable under an exercise assumption, a valid alternative if the reviewed layout has one, and an outside meeting point. A second test blocks both modeled exits and demonstrates an honest no-route result.

8. Native stack and system boundaries

Layer	Recommended implementation
UI	SwiftUI, Observation, native navigation/sheets, SF Symbols, a small design-token layer.
Core	Foundation-only Swift package with Codable models, discrete events, ranking, comparisons, and graph algorithms.
Capture / display	RoomPlan adapters and RealityKit views outside the portable package.
Storage	Actor-owned encrypted Codable vault; Keychain key; complete file protection; atomic writes.
Geographic view	MapKit for SwiftUI with official public layers and sourced resource markers.
Plans	Validated PlanDocument to local UIGraphicsPDFRenderer; PDFKit preview; system sharing/printing.
Optional AI	Explicit on-device SystemLanguageModel, capability checks, validated evidence, deterministic fallback.
Public API	TypeScript/Hono/Zod on Workers; versioned JSON packs; cache/expiry handling; Vitest.

Target iOS 18 for the first build and gate newer features. LiDAR availability and Apple Intelligence availability are independent. Verify exact APIs and concurrency requirements against the actual installed SDK. Do not introduce a second application runtime for an iOS-only product.

The private engine should run without network access. The server supplies public facts; the phone joins them with private household information. A small public-data backend can be prepared before the frontend is polished without creating a server-side household database.

Cloudflare Workers and Hono are reasonable defaults for a small service and local test environment. Begin with static public packs and a few endpoints; add R2 for versioned assets or D1 for genuinely needed public metadata. Verify quotas for deployment and avoid claiming unlimited free infrastructure. No vector database or general agent platform is required. [S30, S31]

9. Maps, regional analysis, and help

Use MapKit as the connected regional display. It provides the requested native Apple appearance while the app adds official hazard polygons and reviewed public resources. Apple's offline Maps downloads are not available to third-party MapKit apps, and cache residency/prefetch cannot be guaranteed. The offline promise therefore covers the home, exercises, plans, selected contacts/addresses, and an app-owned meeting-point diagram. Full regional offline cartography would require separately licensed tiles and additional work. [S13]

Public source	Supported use	Boundary
NWS API	Official alerts, forecasts, observations, zone metadata.	Respect cache/expiry and source attribution; preserve official alert text; no outage prediction.
FEMA flood maps	Effective mapped flood-hazard context and official links.	Not live water depth, road passability, or a parcel safety verdict.
County/city OEM and utilities	Official contacts, enrollment, local advisories and published resources.	An office or map POI is not automatically an active shelter.
Transport operators	Published contacts, service/accessibility information.	Do not assume service continues or a ride is available during a disaster.

The area-analysis agent should use allowlisted public tools and produce typed, source-cited claims. Good questions include which official advisories apply, what flood-map coverage exists, which office serves the region, and whether a cooling or warming resource is actually activated. Combine those facts with household needs locally. Avoid sending a home scan, private address, age bands, equipment notes, or contacts to a remote model. [S17, S18]

Start with a Texas-wide pack and one carefully verified county/urban-area pack; expand across the 13 counties without fabricating coverage. Store source URL, publisher, jurisdiction, source-issued time, last check, expiry, accessibility/pet facts, and unknown status. Keep demo data separate from current information.

Useful help entries include 211 Texas (211 or 877-541-7905), the applicable emergency-management office, water/utility contacts, STEAR, and DisasterAssistance.gov. STEAR registration shares information with responders and does not guarantee assistance; it should be a deliberate external handoff. FEMA aid depends on disaster declaration and program eligibility. [S27-S29]

The backend should accept allowlisted public region identifiers and pack versions only. Household information must not leak in GET parameters, logs, analytics, or arbitrary URLs. Validate incoming content, impose response/time limits, retain a known-good pack when refresh fails, and never show an expired warning as current.

10. Privacy that can be demonstrated

Recommended statement: 'Your household profile and home scan stay on your device. Dry Run does not upload them.' This is more precise than claiming that no one could ever access the information. Intentional exports and third-party maps/registration have separate data boundaries.

Use CryptoKit AES.GCM with a random local key stored in Keychain. For foreground private access, use the WhenUnlockedThisDeviceOnly accessibility class and complete file protection. Use fresh nonces, authenticated envelope metadata, atomic replacement, schema migration guards, and protection attributes on the final files. Private data includes plans, model annotations, and rehearsal histories, not just the initial questionnaire. [S14]

Disable cloud sync initially and exclude the private vault from automatic backup where applicable. Explain that this design does not provide automatic recovery on a replacement phone. A later encrypted backup needs a deliberate user-controlled recovery design; server-held recovery keys would change the promise.

Prevent secondary leaks through raw scan retention, log output, crash attachments, app-switcher previews, PDF metadata/filenames, share temporaries, widgets, indexing, or sensitive notifications. Do not promise to prevent every screenshot. Export only the chosen plan, and clean up unneeded plaintext files. A delete-all flow must include derived private caches and keys while preserving unrelated public content.

A privacy test should inject a fictional canary into every sensitive field and inspect network calls, local public files, logs, exports, and lifecycle transitions. Document which behaviors were actually tested on physical hardware. Encryption-library selection alone does not establish a complete privacy guarantee.

11. LLMs and multilingual content

Two AI roles are useful: a remote/public regional researcher working only with public sources, and an optional on-device explanation layer for already computed household results. Neither owns emergency decisions, route selection, arithmetic, medical compatibility, priority ordering, or authoritative instructions.

Use SystemLanguageModel explicitly for the personal layer and check runtime availability. The broader Foundation Models framework can expose more than one execution/provider option, so the framework name alone is not a privacy guarantee. Refusal, unsupported language, unavailable model, timeout, or invalid output should immediately fall back to a deterministic explanation. Do not circumvent model safeguards. [S15, S16]

Give the model a small evidence bundle. Valid action/source IDs help but do not prove that generated wording preserved a prohibition or quantity. Keep actionable instructions in reviewed templates and restrict generated prose to noncritical explanations of computed consequences. The app should feel complete when AI is off.

Support English, Spanish, and Vietnamese through versioned locale resources with source/review metadata. Add additional languages when their instructions are reviewed. Official translated materials can seed appropriate portions, but do not assume a translated source covers every scenario. Treat model-generated translations as drafts. Preserve numerical values, conditions, negation, contacts, units, and visible fallback language. If unreviewed safety text is previewed or exported, visibly label it 'Draft translation - language review pending' so that printed copies retain their review status.

Produce a household essentials sheet and a one-page plan per selected scenario. Include immediate steps, the relevant home/shelter/meeting diagram, assistance and transport arrangements, essential equipment notes, top preparations, offline contacts, and review/content dates. Keep medical details minimal by default. Use readable typography and intentional prioritization rather than shrinking every line to fit.

The plan viewer and PDF creation work offline. Printing requires an available printer connection. A QR code can supplement, but cannot replace, printed phone numbers and essential address text.

12. Interface direction

Use four tabs: Rehearse, Home, Plan, and Help. Put progressive setup beside the Home model. A fictional household demo bypasses onboarding. Keep a focused next action on each screen rather than presenting many status tiles.

Rehearse should combine the visible exercise state, a controllable clock, a concise event sequence, and a decision card. The debrief shows three high-priority actions and a replay comparison. Home exposes the dollhouse, 2D plan, floor selection, and confirmations. Plan holds language selection, saved pages, and real preparation status. Help combines a saved directory with the optional online map.

Visual defaults: warm neutral surfaces, near-black type, restrained blue or teal accents, generous spacing, and clear amber/red status labels. Use native controls and simple symbols. Reserve prominent color for meaning. Critical instructions belong on opaque, legible surfaces rather than over animated 3D content.

Support Dynamic Type, VoiceOver, Reduce Motion, Reduce Transparency, and non-color state labels. Keep a first-class accessible room list. Do not use sirens by default or game mechanics that reward delaying escape, searching for objects, or taking risks to gain a score.

Run graph work, parsing, simulation, and PDF composition with appropriate background isolation. The clock advances simulated time independently of rendering frames. Reuse scene entities, cancel outdated tasks, and discard results from stale revisions. Pause unnecessary 3D work offscreen and offer a 2D low-power path. Device performance remains a measurement, not an assumption.

13. Three-hour implementation plan

Use a lead integrator and three subagents. The lead owns contracts, project files, the thin native shell, integration, and later UI/PDF work. One agent owns simulation/ranking; another home geometry/capture/routing; the third privacy/public data/content. Freeze shared interfaces early and allocate disjoint files or worktrees.

Window	Concrete objective
0-15 minutes	Inspect repo/environment; establish package/app/backend targets, contract ownership, fixtures, and acceptance criteria.
15-60 minutes	Implement freeze reference engine, home portal graph, vault, and public pack contract in parallel; integrate a thin runnable slice.
60-115 minutes	Complete six distinct rule packs, evidence trace, ranking/replay, routing constraints, cache/directory, and persistence.
115-150 minutes	Connect the native journey, plans/locales, and map directory; have a finished agent review another lane.
150-160 minutes	Repair high-impact integration findings and exercise offline/unsupported-device cases.
160-180 minutes	Freeze features, verify, fix concrete failures, commit coherent work, and prepare the first-review dossier.

Core and backend receive the early effort. A thin shell exists early to catch interface mismatches; visual refinement waits until the data flow works. Optional AI and broad geographic coverage should not block a useful product. The handoff should preserve incomplete requirements in a precise backlog instead of disguising them as completed features.

Three hours can produce a substantial first review. It cannot prove real-home reconstruction accuracy, production privacy guarantees, correct translations for every emergency, and reliable device behavior. Native SwiftUI/RoomPlan compilation requires an appropriate Xcode environment; actual scanning requires compatible hardware. On Linux, code and backend/core tests can still advance while native validation remains explicitly pending.

The review package should contain source code, run instructions, a five-minute demo path, architecture/decision notes, source manifests, a verification matrix, an AI-assistance ledger, and the next student-owned technical work. The separate Codex handoff specifies detailed models, module ownership, algorithms, sources, tests, and expected deliverables.

14. Acceptance and the next review

Area	Evidence required
Cascades	Correct conditional dependencies; shared resources conserved; stable event ordering; unknowns preserved; restoration does not reset unrelated failures.
Counterfactuals	Same external events and initial state; intervention isolated; removing it reproduces the baseline.
Fire routes	Valid geometric polylines; blocked exits excluded; mobility constraints enforced; no invented windows/exits; shared bottlenecks disclosed.
Privacy	No private canary in public requests/logs; ciphertext tampering handled; atomic saves; deliberate exports; lifecycle/deletion checks.
Offline/content	All six rehearsals and plans work without network or AI; stale/demo alerts never appear live; locale/PDF correctness inspected.
Native integration	Actual build/simulator/device status recorded separately; unsupported capabilities have working fallbacks.

For the first demonstration, use a fictional household with electricity-dependent heat, a well, and essential powered equipment. Show the freeze cascade, explain one important dependency, add a verified preparation, and rerun to show the specific change. Then show the home-fire blocked-route exercise and a saved plan in airplane mode. This makes the product's contribution visible without touring every screen.

The next review should decide whether onboarding asks the right questions, whether families understand the cascade explanations, whether the home confirmation step is manageable, whether alternate-route limits are clear, and whether the action list feels useful for households with little money or no car. Frontend refinement should follow those findings.

Before public release, complete real-device RoomPlan and performance checks, practical home-model validation, fire/emergency-content review, medical-equipment boundary review, language review, and a privacy data-flow audit. These are concrete validation needs of the promised features. They should not prevent building and inspecting the reference prototype now.

Sources and implementation references

Source IDs in this brief match the standalone Codex handoff. Click a source title to open the reference. Research was checked on October 9, 2026; current services, rules, and SDK details should be rechecked during implementation. Engineering recommendations are distinguished from source claims throughout.

S01 Congressional App Challenge, 2026 official rules

S02 Congressional App Challenge registration/deadline

S03 Texas 2021 enacted district description, Section 25; current geography must still be checked for submission

S04 Census Reporter, TX-25, ACS 2024 one-year estimate

S05 Texas DSHS final February 2021 winter-storm mortality report

S06 Texas A&M, state climatologist's 2024 assessment through 2036

S07 Official DisasterReady winner profile

S08 Apple RoomPlan overview

S09 Apple multiroom workflow and published recommendations

S10 RoomCaptureSession.isSupported

S11 RealityView

S12 Apple SceneKit-to-RealityKit guidance

S13 Apple DTS, MapKit offline limitations

S14 Apple Keychain data protection

S15 Apple on-device SystemLanguageModel

S16 Foundation Models acceptable-use requirements

S17 NWS API documentation

S18 FEMA National Flood Hazard Layer

S19 USFA home-fire escape plans

S20 CDC power-outage safety

S21 CDC heat-health guidance

S22 NWS tornado response

S23 NWS flood response

S24 CDC drinking-water advisories

S25 FDA medical devices and disasters

S26 CDC emergency water storage

S27 211 Texas contact information

S28 TDEM STEAR

S29 USAGov disaster-assistance eligibility

S30 Hono on Cloudflare Workers

S31 Cloudflare Workers limits